

Test Sample Paper A
With Answers and Workings

Total Score
100



Name: _____

Section: _____ Classes held on: _____

INSTRUCTIONS TO STUDENTS

1. Write your full name, section number and timing of your classes in the space provided above.
2. This paper contains **2** sections. Section A has 12 questions of 3 marks each. Section B contains questions 13-30 and the marks are stated next to the each question.
3. You are required to answer **ALL** questions. Write your answer in the space provided after each question. If applicable, write your workings legibly in the space provided after each question. No separate papers are allowed.
4. For multiple choice questions, choose the **MOST APPROPRIATE** answer from the alternatives given. Multiple answers are **not** applicable for this paper.
5. This test constitutes 20% of your total assessment of this MA module.
6. Time allowed is 90 minutes

Suggestion: Check your final answers using this answer key and return to workings if you find a mismatch. Try to detect your own error before looking at the full answers in the rest of the document.

Answer Key	
1	E
2	C
3	A
4	A
5	A
6	E
7	B
8	E
9	B
10	B
11	C
12	D
13	\$215,000
14	\$200,000 UNDER-APPLIED
15	16,000
16	5X, 20%
17	\$6,838,400
18	\$4,410,400
19	300,000 EUs
20	\$3.910
21	\$184,800
22	\$270,000
23	\$262,500
24	\$23,200
25	\$22,760
26	\$606,440
27	\$2,500 per batch
	\$2,000 per inspection
	\$800 per mm
	\$1,000 per inspection
28	\$159,500
29	\$78,500

1. Anron Manufacturing produces a product which has 3 different costs and a relevant range between 2,500 units and 10,000 units per month. Per-unit costs at two different activity levels for each cost are presented below.

<u>Activity</u>	<u>Cost A</u>	<u>Cost B</u>	<u>Cost C</u>	<u>Total Cost</u>
5,000 units	\$4 per unit	\$10 per unit	\$4 per unit	\$18 per unit
8,500 units	\$4 per unit	\$8 per unit	\$2.50 per unit	\$14.50 per unit

The cost formula that expresses the behavior of Anron's total cost is:

- A) $Y = \$0 + \$18X$
- B) $Y = \$20,000 + \$14.5X$
- C) $Y = \$40,000 + \$10X$
- D) $Y = \$50,000 + \$4X$
- E) $Y = \$42,500 + \$9.5X$

Answer: [_____]

Calculate total costs for 5,000 units and 8,500 units and use the high low method.

	<u>Units</u>	<u>Total Costs</u>
Low	5,000	$5,000 \times \$18 = \$90,000$
High	8,500	$8,500 \times \$14.5 = \$123,250$
Difference	3,500	\$33,250

$VC \text{ per unit} = \$33,250 / 3,500 = \9.5 , $Total \text{ FC} = \$90,000 - (5,000 \times \$9.5) = \$42,500$

2. Based on an account analysis of the security costs for Security Jones Company (SJC), the security costs was \$12.50 per labor-hour at an activity level of 160 labor hours and \$10 per labor-hour at an activity level of 200 labor-hours. Assuming the activity levels are within the relevant range, the total expected security cost if the activity level is 180 labor-hours would be:

- A) \$4,050
- B) \$2,250
- C) \$2,000
- D) \$2,025
- E) \$1,800

Answer: [_____]

$\$12.5 \times 160 = \$2,000$
 $\$10 \times 200 = \$2,000$

3. After the breakeven point, ceteris paribus (i.e. with other conditions remaining the same), which of the following is false

- A) Degree of Operating Leverage (DoL) will have a greater value with the increase in sales.
- B) Increase in contribution margin as a result of additional sales will increase net operating income by the same amount of the increase in contribution margin.
- C) Margin of safety ratio will keep increasing as sales increase.
- D) The breakeven ratio will get smaller as sales increase.
- E) None of the above.

Use hypothetical figures to do this question is easier!

4. Angie's Electrical Store produces washers and dryers in an assembly-line process. Labor costs last month were as follows:

- corporate executives, \$100,000
- assembly-line workers, \$80,000
- janitors, \$22,000
- plant supervisor, \$40,000.

The total direct labor cost was

- A) \$ 80,000
- B) \$102,000
- C) \$120,000
- D) \$142,000
- E) \$242,000

Answer: [_____]

5. The accounting records of XYZ Company that makes many different products revealed the following costs, among others:

Factory insurance	\$ 22,000
Direct material used	\$320,000
Sales commission	\$ 25,000
Indirect labor	\$ 32,000
Depreciation of sales persons' cars	\$ 18,000
Production equipment rental costs	\$ 64,000

The total manufacturing overhead were:

- A) \$118,000
- B) \$136,000
- C) \$86,000
- D) \$481,000
- E) None of the above

Answer: [_____]

Factory insurance	\$ 22,000
Indirect labor	\$ 32,000
Production equipment rental costs	\$ 64,000
Total	\$118,000

6. Assuming you accidentally overstated the quarter's depreciation expense for office equipment by \$45,000. Which of the following correctly depicts the effect of this error?

- A) Cost of goods manufactured will be overstated by \$45,000.
- B) Cost of goods sold will be overstated by \$45,000.
- C) Both cost of goods manufactured and cost of goods sold will be overstated by \$45,000.
- D) Cost of goods sold will be overstated by \$45,000, and cost of goods manufactured will be understated by \$45,000.
- E) None of these. (office equipment is to do with selling and administration expenses)**

Answer: [_____]

7. The contribution-margin format of the income statement:

- A) is used with absorption costing
- B) highlights the lump sum of fixed manufacturing costs**
- C) distinguishes manufacturing costs from nonmanufacturing costs
- D) calculates gross margin
- E) is used for external reporting purpose.

Answer: [_____]

8. Which of the following statements relating to variable costing and absorption costing is true?

- A) The product costs for absorption costing differ from that of variable costing in that the product costs for absorption costing includes variable manufacturing overhead.
- B) In the absence of any fixed costs, contribution margin will always be the same as gross profit.
- C) In using absorption costing to cost a product, one must include all fixed costs, manufacturing and non-manufacturing, in the computation of the product cost.
- D) Contribution margin is always equal to sales minus all variable manufacturing costs.
- E) None of the above.**

Answer: [_____]

9. For performance evaluation purposes, the fixed costs of a service department should be charged to operating departments using:

- I. actual fixed costs and the budgeted level of activity for the period.
- II. budgeted fixed costs and the budgeted peak-period/long-run average servicing capacity.
- III. actual fixed costs and the actual peak-period/long-run average servicing capacity.

A. I only.

B. II only.

C. III only.

D. I and II only.

E. II and III only.

Answer: [_____]

10. The clerical activity associated with preparing purchase orders to buy materials to produce a production run of a standard product is an example of a:

A. Unit-level activity. *(Incurred for each unit of the product)*

B. Batch-level activity. *(Each production run is a batch. One purchase order for one production run.) Ordering materials is a batch level activity as the ordering process is done by batches. Orders are made to cope with short-term requirements for up to an expected units or batches of productions)*

C. Product-level activity.

D. Organization-sustaining activity.

E. Customer-level activity.

Answer: [_____]

11. Which of the following statements is **not** true?

A. In a job-costing system, individual jobs use different quantities of production resources.

B. In a process-costing system each unit uses approximately the same amount of resources.

C. In the weighted-average process costing method, the costs of direct materials in beginning inventory are not included in the equivalent unit cost calculation since direct materials are almost always added at the start of the production process.

D. Examples of industries that would use process costing include the pharmaceutical and soft drinks manufacturing industries.

E. In process costing calculation, conversion costs are usually assumed to be added evenly throughout the production process

Answer: [_____]

12. When considering the appropriateness of costing technique(s) e.g. job-order costing, process costing, activity-based costing, to be adopted for internal decision making purpose, which of the following is the **least important** or **irrelevant** factor to a company?

A. The techniques should match and fit the underlying operation involved

B. The techniques should be simple and cost effective

C. The company should have sufficient resources and manpower to run the techniques/system

D. The techniques should be IFRS/GAAP compliant

E. The techniques should reflect consumption of resources by the cost objects

Answer: [_____]

SECTION B: Structured Questions (64 marks total)

13. Ace Company Singapore (ACS) reported Cost of Goods Sold of \$330,000 for the February 2012. The beginning Work-in Process Inventory was \$140,000 and the ending Work-in Process Inventory was \$35,000. Beginning Finished Goods Inventory was \$170,000 and the ending Finished Goods Inventory on was \$55,000. What was the amount of Cost of Goods Manufactured for the year?

Answer: [_____]
(2 marks)

MUST SHOW WORKINGS

Beg Finished Goods	\$170,000
Cost of Goods Manufactured	\$215,000
End Finished Goods	(\$55,000)
Cost of Goods Sold	\$330,000

14. The management of MA Corporation would like to investigate the possibility of basing its predetermined overhead rate on activity at capacity. The company's financial controller has provided an example to illustrate how this new system would work. In this example, the allocation base is machine-hours.

	Budgeted	Capacity	Actual
Machine-hours	48,000	60,000	50,000
Manufacturing overhead	\$1,200,000	\$1,200,000	\$1,200,000

If the company bases its predetermined overhead rate on capacity, by how much was manufacturing overhead underapplied or overapplied?

Answer: [Under or Over-applied (please delete one)

_____]
(3 marks)

MUST SHOW WORKINGS

POR = Bgt MOH \$1,200,000/Capacity 60,000 MH = \$20 per MH

Applied MOH = \$20 x Actual 50,000 MH = \$1,000,000

Underapplied MOH = Applied MOH 1,000,000 – Actual \$1,200,000 = \$200,000

Use the following information for Questions 15-16:

Ben and Co sells four different products and has the following information for last month:

	A	B	C	D	Total
<i>Units Sold</i>	<i>20,000</i>	<i>50,000</i>	<i>80,000</i>	<i>10,000</i>	<i>160,000</i>
Sales	\$230,000	\$100,000	\$150,000	\$30,000	
Variable Expenses	\$184,000	\$75,000	\$90,000	\$15,000	
Contribution Margin	\$46,000	\$25,000	\$60,000	\$15,000	\$146,000
Fixed Cost					\$116,800
Operating Income					\$29,200

15. Compute the break-even units for Product A by using last month's unit sales mix.

Answer: [_____]
(4 marks)

16. Compute the company's degree of operating leverage and margin of safety percentage for last month.

Answer: [_____]
(4 marks)

MUST SHOW WORKINGS

Q15

	A	B	C	D	Total
<i>Units Sold</i>	<i>20,000</i>	<i>50,000</i>	<i>80,000</i>	<i>10,000</i>	<i>160,000</i>
Sales	\$230,000	\$100,000	\$150,000	\$30,000	
Variable Expenses	\$184,000	\$75,000	\$90,000	\$15,000	
Contribution Margin	\$46,000	\$25,000	\$60,000	\$15,000	\$146,000
Fixed Cost					\$116,800
Operating Income					\$29,200
Unit CM	\$2.300000	\$0.500000	\$0.750000	\$1.500000	\$0.912500
Weighted unit CM	\$0.287500	\$0.156250	\$0.375000	\$0.093750	\$0.912500
BE Units	**16,000				*128,000

Method 1: Using Weighted Unit CM

* $\$116,800 / \$0.9125 = 128,000$

** $20,000 / 160,000 * 128,000 = 16,000$

Method 2: Using BE%

** $BE\% = 116,800 / 146,000 = 80\%$. Qty of Product A for company to breakeven = $20,000 * 80\% = 16,000$ units

Q16

DOL = $CM / \text{Operating Income} = 146,000 / 116,800 = 5$ times

$$\text{MOS\%} = 1/\text{DOL} = 20\%; \text{ OR } 1 - \text{BE\%} = 1 - 80\% = 20\%$$

Use the following information for Questions 17-18:

Pang and Company manufactures a special self-balancing zero emissions personal transportation vehicle, the vehicle has two wheels and takes up very little space. It can be used for day to day transportation as well as for recreation. The company uses actual costing. Its actual data for August were:

Selling price	\$3,000
Units in beginning inventory	600
Units produced	3,000
Units sold	3,200
Units in ending inventory	400
Variable manufacturing costs per unit	\$830
Variable selling and administrative costs per unit	\$33
Total fixed manufacturing overhead	\$1,800,000
Total fixed selling and administrative expenses	\$508,000

17. Compute the contribution margin for August.

Answer: [_____]
(3 marks)

18. Compute the net operating income using the variable costing format and use this figure to determine the absorption costing net operating income for August. Assume July's fixed manufacturing overhead per unit is the same as August's.

Answer: [_____]
(4 marks)

MUST SHOW WORKINGS

Variable costing income statement		
Sales		\$9,600,000
Beginning inventory	\$498,000	
Variable manufacturing costs	\$2,490,000	
Ending inventory	(\$332,000)	
Cost of goods sold		(\$2,656,000)
		\$6,944,000
Variable selling and administrative costs		(\$105,600)
Contribution margin		\$6,838,400
Total fixed manufacturing overhead		(\$1,800,000)
Total fixed selling and administrative expenses		(\$508,000)
Variable costing profit		\$4,530,400
Absorption costing profit		
Variable costing profit		\$4,530,400
Change in inventory * unit FMOH =		(\$120,000)
Absorption costing profit		\$4,410,400

Use the following information for questions 19-21:

Trophy Pet Food Limited is a manufacturer and supplier of a special beef flavor dog food in cans. The company has three departments to manufacture the can food for dogs. The first department is the mixing of the raw food ingredients and then these mixed ingredients are transferred to the second department for cooking. After the cooking process is complete, the cooked ingredients are transferred to the final department for canning.

Cooking Department

In the cooking department, a beef flavor extracted from tissues of cattle is added when the cooking process is 30% completed and seasonings are added when the cooking process is 90% complete. Conversion costs are assumed to be incurred uniformly throughout the entire production process.

The following is a summary of the May 20x3 operations of the cooking department:

	Physical Units (no. of cans)
Work-in-process as at 1 May (20% completed)	90,000
Transferred in from the mixing department	270,000
Work-in-process as at 31 May (40% completed)	60,000

For the work-in-process units as at 1 May, \$182,000 and \$23,760 were incurred regarding transferred-in cost and conversion costs respectively.

During May, the mixing department transferred \$574,000 mixed ingredients into the cooking department.

Other costs incurred in May for the cooking department include beef flavor of \$165,600, seasonings of \$15,000 and conversion costs of \$397,440.

The company uses the weighted average cost method.

Show your workings on next page.

19. How many equivalent units are to be accounted for during May 20x3 with respect to seasonings?

Answer: [_____]
(3 marks)

20. What is the total unit cost per can for the month of May 20x3?

Answer: [_____]
(3 marks)

21. What is the cost of ending work-in-process at the end of May 20x3?

Answer: [_____]
(3 marks)

Workings for Questions 19-21**WA**

1. PU						
BWIP	90,000					
Started	270,000					
Units to account for	360,000					
		TIC	Beef Flavor	2. EU Seasonings	CC	
Completed	300,000	300,000	300,000	300,000	300,000	
EWIP	60,000	60,000	60,000	0	24,000	
Units to account for	360,000	360,000	360,000	300,000	324,000	
3. Unit Costs		TIC	Beef Flavor	Seasonings	CC	Total
BWIP		\$182,000	\$0	\$0	\$23,760	\$205,760
Incurred		\$574,000	\$165,600	\$15,000	\$397,440	\$1,152,040
		\$756,000	\$165,600	\$15,000	\$421,200	\$1,357,800
Unit Costs		\$2.100	\$0.460	\$0.050	\$1.300	\$3.910
4. Total Costs						
Completed						\$1,173,000
EWIP		\$126,000	\$27,600	\$0	\$31,200	\$184,800
Total						\$1,357,800

Use the following information for Questions 22-23:

Intello is a computer software consulting company that provides three major services to clients i.e. Computer Programming, Consulting and Training. The following data pertains to its budget for the coming year:

	Service Departments		User Departments/Profit Centres		
	Info System	Facilities	Programming	Consulting	Training
Budgeted Hours chargeable to clients			10,000 hrs	20,000 hrs	15,000 hrs
Budgeted Overhead before allocation	\$150,000	\$80,000	\$175,000	\$210,000	\$185,000
Computer Hours used by Dept		650	1,300	600	700
Floor space of Dept (sqm)	200		400	500	700

22. Compute the total budgeted overhead costs of the **Programming Dept.** if the service departments' overheads are allocated to the user departments using the **direct method**. Allocate the Info System cost on the basis of computer hours used and the Facilities cost on the basis of Floor Space. **Answers for Consulting and Training departments are not required.**

Answer: [_____]
(4 marks)

$$\text{\$175,000} + \text{\$150,000} \times \frac{1,300}{(1,300+600+700)} + \text{\$80,000} \times \frac{400}{(400+500+700)} = \text{\$270,000}$$

23. Compute the total budgeted overhead costs of the **Programming Dept.** if the service departments' overheads are allocated to the user departments using the **step-down method** starting from Info System. **Answers for Consulting and Training departments are not required.**

Answer: [\$ _____]
(4 marks)

Show your working

	Info System	Facilities	Programming	Consulting	Training	Total
Budgeted Overhead before allocation	\$150,000	\$80,000	\$175,000	\$210,000	\$185,000	\$800,000
(1) Allocate from IS to Facilities & Users)						
Computer Hours		650	1,300	600	700	3,250
Proportion (Computer Hrs)		0.200	0.400	0.185	0.215	
Allocation from IS Syst	- \$150,000	\$30,000.00	\$60,000	\$27,692.31	\$32,307.69	
Bgt OH after allocation from IS Syst	\$0	\$110,000	\$235,000	\$237,692	\$217,308	
(2) Allocate from Facilities to Users						
Floor space (sqm)			400	500	700	1,600
Proportion (sqm)			0.250	0.313	0.438	
Allocation from Facilities		-\$110,000	\$27,500	\$34,375.00	\$48,125.00	
Bgt OH after allocation from IS Syst & Facilities	\$0	\$0	\$262,500	\$272,067.31	\$265,432.69	\$800,000

Use the following information for Questions 24-30:

ABC Ltd uses a **normal** costing system. On 1 January, it identified the following budgeted overhead costs and cost drivers for the coming year:

Overhead Activities	Activity Driver	Budgeted Cost
Machine setup	No. of batches	\$20,000
Inspection	No. of inspections	\$130,000
Material handling	No. of material moves	\$80,000
Engineering	Engineering hours	\$50,000
Total		\$280,000

Total budgeted direct labour for the company for the year was **20,000 hours (\$100,000)**.

The company started three jobs during the year:

	Job 101	Job 102	Job 103
Status of Job as at 31 Dec	Completed	In Progress	Completed and delivered to customers
Direct materials	\$5,000	\$250,000	\$8,000
Direct labour	\$2,000	\$93,800	\$4,000
Units started in production	100	5,000	200
Batch size for each machine set up	100 units	2,500 units	40 units
Number of inspections	20	0	45
Number of material moves	30	20	50
Engineering hours	5	40	5

Assume actual direct labour cost per hour is the same as the budget; and actual quantities of the activity drivers are the same as the budget.

On 31 December, ABC Ltd noted that actual total overhead for the year amounted to \$279,000.

If ABC Ltd uses a **plant-wide rate** (for Questions 24-26) to allocate overhead based on direct labour hours.

24. Compute the cost of **Job 103**.

Answer: [_____]
(4 marks)

Total Bgt Overhead	
=	\$280,000
Total Bgt DLH =	20,000
POHR =	\$14 per DLH
DM	\$8,000
DL	\$4,000
Applied OH	\$14 * (4000/5##) = \$11,200
Cost of Job 103	\$23,200

5 = \$100,000/20,000 rate per hour

25. If ABC Ltd writes off under- or over-applied overhead to the cost of goods sold, compute the **adjusted** cost of goods sold.

Answer: [_____]
(4 marks)

Total Applied Overhead =	$\$14 * [(\$2,000 + \$93,800 + \$4,000) / 5]$	
Total Actual Overhead =		\$279,440
Overapplied Overhead =		<u>\$279,000</u>
		<u>\$440</u>
Only <u>Job 103</u> is sold; thus		
Adjusted COGS =	$\$23,200 - \$440 =$	<u>\$22,760</u>

26. Compute the cost of WIP inventory as at 31 December.

Answer: [_____]
(4 marks)

WIP as at 31 Dec from <u>Job 102</u> only		
DM		\$250,000
DL		93,800
Applied OH $\$14 * \$93,800 / \$5$		<u>262,640</u>
Cost of Job 102		<u>\$606,440</u>

If ABC Ltd uses **activity-based costing (for Questions 28-30)**

27. Compute the activity rates (fill in the blanks)

(4 marks)

Overhead Activities	Cost Driver	Budgeted Cost	Activity Level	Activity Rate
Machine setup	No. of batches	\$20,000	$100/100 + 5,000/2,500 + 200/40 = 8$	\$2,500 per batch
Inspection	No. of inspections	\$130,000	65	\$2,000 per inspection
Material handling	No. of material moves	\$80,000	100	\$800 per mm
Engineering	Engineering hours	\$50,000	50	\$1,000 per eng hr

28. Compute the cost of goods sold for the year **before** adjusting for under-/over-applied overhead.

Answer: [_____]

(4 marks)

Job 103			
DM			\$8,000
DL			\$4,000
Applied OH:			
Machine setup	$\$2,500 * 5 =$	\$12,500	
Inspection	$\$2,000 * 45 =$	\$90,000	
Material handling	$\$800 * 50 =$	\$40,000	
Engineering	$\$1,000 * 5 =$	\$5,000	
Total Applied OH			\$147,500
COGS			\$159,500

29. Compute the dollar amount of finished goods inventory as at 31 December.

Answer: [_____]
(4 marks)

Finished goods as at 31 Dec from Job101 only

Job 101

DM **\$5,000**

DL **\$2,000**

Applied OH:

Machine setup $\$2,500 * 1 =$ **\$2,500**

Inspection $\$2,000 * 20 =$ **\$40,000**

Material handling $\$800 * 30 =$ **\$24,000**

Engineering $\$1,000 * 5 =$ **\$5,000**

Total Applied OH **\$71,500**

Cost of FG @ 31 Dec: Job 101 **\$78,500**

30. Explain the main reason for the difference in the cost of goods sold computed using a plant-wide rate and activity-based costing. Ignore the adjustments for over- or under-applied overhead for both methods.

(3 marks)

Write your answer within the box:

- Traditional and ABC costing systems generally result in comparable product costs when overhead is a small portion of costs. However, for ABC Ltd, overhead is about 40% of total manufacturing costs.
- Traditional costing allocates overhead based on direct labour. As Job 103 consumes a small proportion of the total DL (about 4%), it is allocated a small portion of the overhead accordingly.

Under ABC, overhead costs are found to be generated from non-unit level or non-direct-labour related activities. Thus DL would not be a reasonable cost allocation base to allocate the overhead. E.g. almost 50% of the overhead comes from Inspection. Inspection cost is driven by the no. of inspections and Job 103's share is about 60% of the total inspections. Thus Job 103 should bear a higher portion of the overhead.

(Most students discussed the differences between the two methods in general and did not refer back to the data in the questions. Students who referred to the actual data got full credit.)

End of the Test